Exercise on Mutation Testing

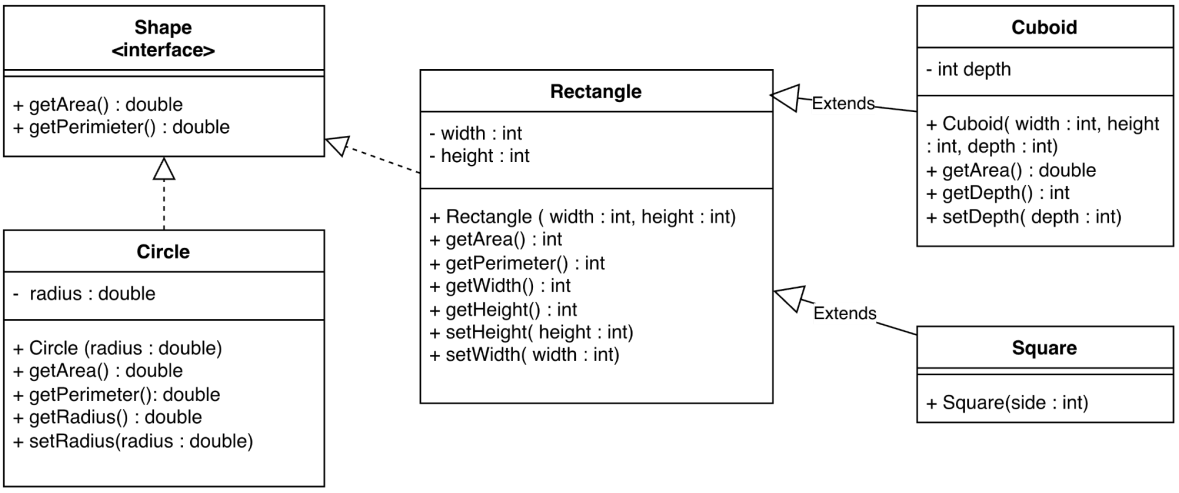
Week 5: Mutation Testing

Setup the project:

1. Install Maven. Search Youtube for guides in your preferred OS.
    a. Make sure your environmental variables are defined.
    b. Run mvn -v on your terminal to see that Maven is working.
2. Download and unzip “QAT_Week5_Exercise_Mutation” from Canvas.
3. Import the Java project to your IDE (e.g., Visual Studio Code)

Part 1: Understanding the Code and Designing Tests

1. The diagram below shows the different classes and their relationships



2. Generate 10 test cases for the classes below.

3. Manually create 10 mutants and add them to the table below:

ID	Mutation Operator	Class	LoC	Mutant
M1				
M2				
M3				
M4				

M5				
M6				
M7				
M8				
M9				
M10				

4. Run your test cases against the mutants. Fill in the mutation matrix and calculate the mutation score.

	M1	**M2**	**M3**	**M4**	**M5**	**M6**	**M7**	**M8**	**M9**	**M10**
T1										
T2										
T3										
T4										
T5	x									
T6										
T7										
T8										
T9										
T10										

5. Write down your reflections on the points below:
    a. Did you use a variety of operators? Were some harder/easier to apply than others?
    b. Was there a type of mutant that was harder / easier to kill?
    c. Was there a test case that is very good at killing multiple mutants?

Part 2: Automating Mutation Testing with PIT

1. PIT should be configured in your project's pom.xml file.
2. In the terminal (project root), run the command (https://pitest.org/quickstart/maven/):

```
mvn test-compile org.pitest:pitest-maven:mutationCoverage
```

3. Check the mutation report in folder: target > pit-reports > index.html

Reflect on the questions:

1. How similar / different were your code and mutation coverage across different classes?
2. Was there a mutation operator from PIT that was harder to kill?
3. Repeat the steps in Part 1 to create new tests to kill the surviving mutants.